

USD/CRC · MONEX — Pricing-Model Reverse-Engineering & Trading Report

BCCR MONEX daily data · 362 trading days · 6 Dec 2024 → 19 Jun 2026 · open/close/low/high · VWAP · volume · trade counts

The USD/CRC is **not a random walk** — it is a **VWAP-anchored crawling managed float** in a strong colón-appreciation trend. **Volume barely affects how big a move is, but strongly predicts its direction**: heavy days are USD-supply days that push the colón stronger, and that pressure **persists into the next session** — the basis of the best tradeable edge.

HEADLINE NUMBERS

−10.1%

colón appreciation

504.99 → 454.10 ¢ over the sample

39 \$M

avg daily volume

median 35 \$M · 216 trades/day

−0.45

bps move per \$M

signed move vs volume · $p \approx 1e-13$

+0.37

lag-1 autocorr

session moves trend, don't revert

Sharpe 4.4

order-flow signal

+7 bps/day · 59% hit (gross)

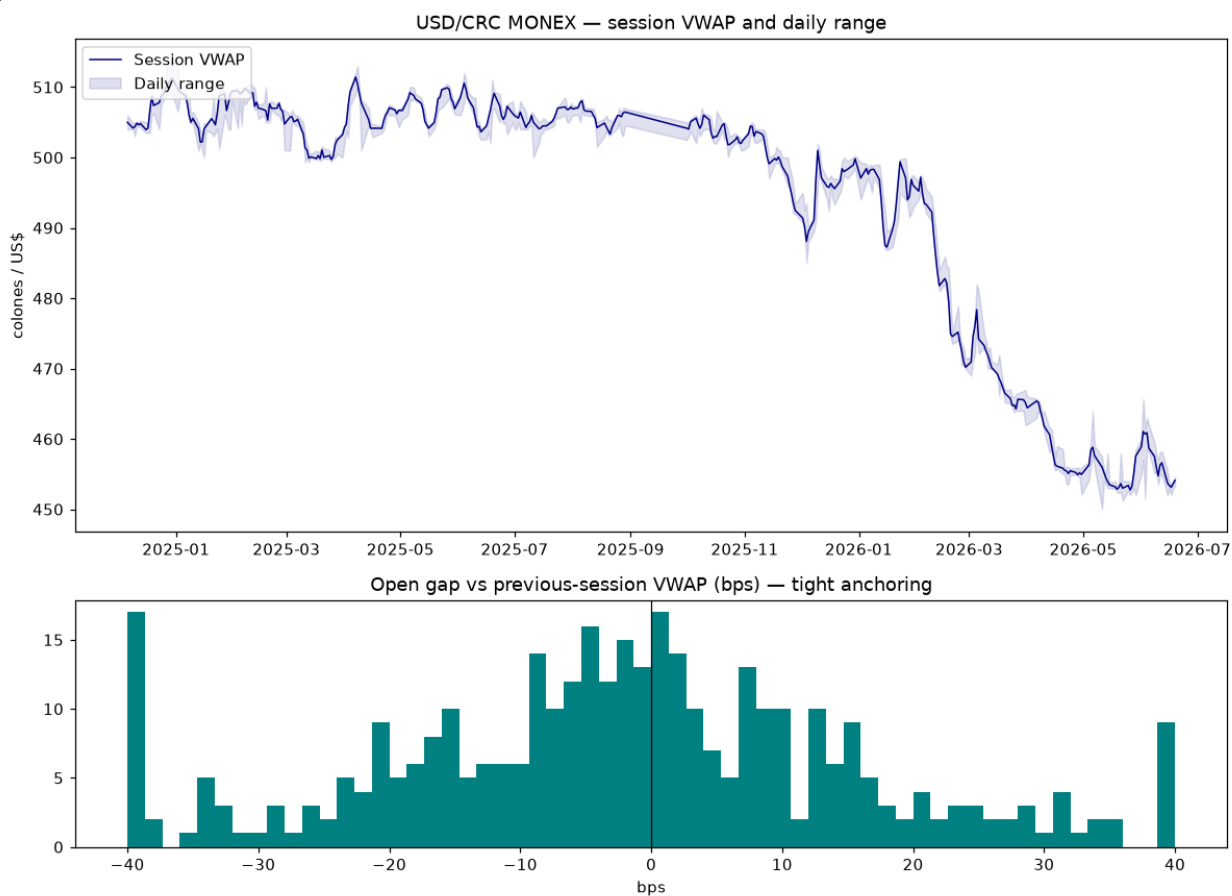
507 ¢

point of control

value area 453–510 ¢

1 · PRICE FORMATION

Session VWAP, daily range & opening-gap distribution



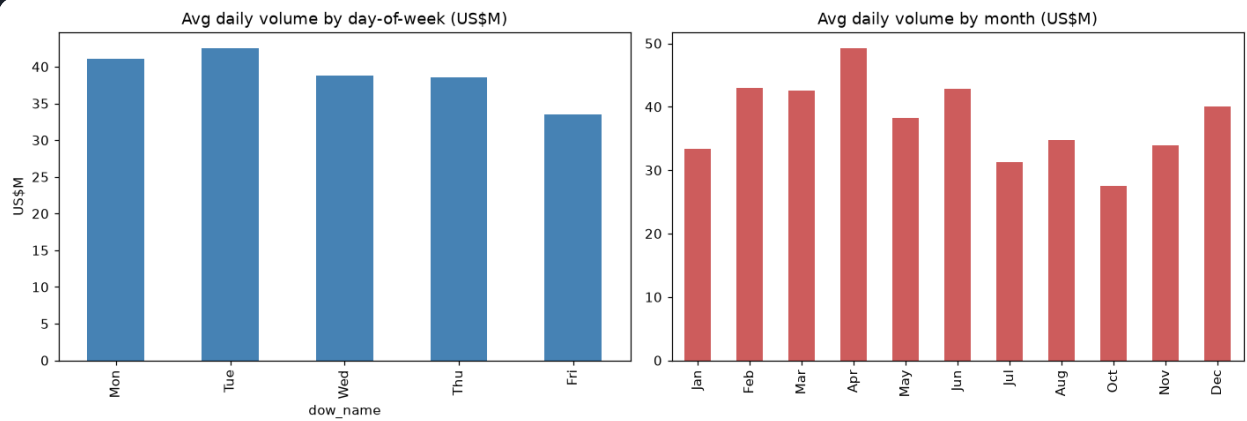
Top: the colón was range-bound ~500–511 through Oct-2025, then appreciated sharply to ~454. Bottom: the open clusters tightly around the prior session's VWAP (median gap -1.6 bps) — the market re-anchors to yesterday's weighted average each day.

Anchor. 48% of sessions open within ± 10 bps of the prior VWAP; ~80% of any opening gap is retraced by the close.

Crawl. Average drift ≈ -37 ¢/yr ($R^2=0.70$), but the regime is "flat $\rightarrow$ steep down", not constant speed.

2 · VOLUME SEASONALITY

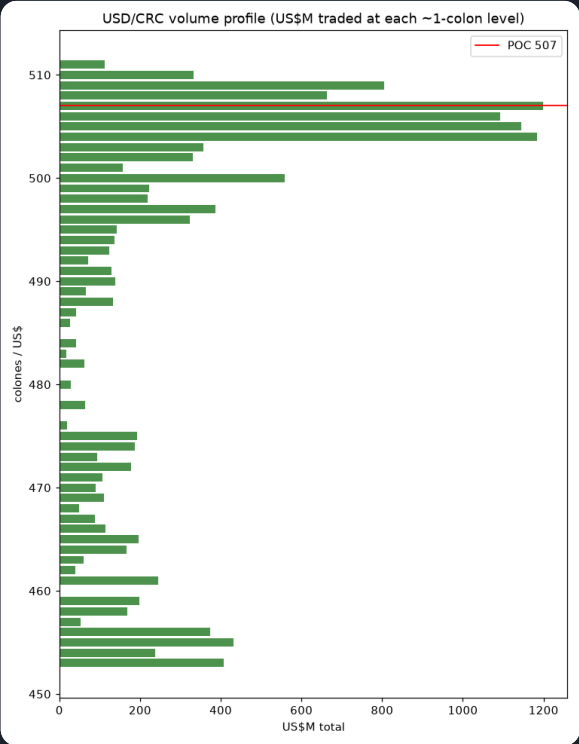
Average daily volume by day-of-week and month



Friday is the lightest day with the smallest tickets (clean-up flow). April is the heaviest month (tax/repatriation); the first half of each month and Feb show the strongest appreciation; December is the only depreciation month (aguinaldo import demand).

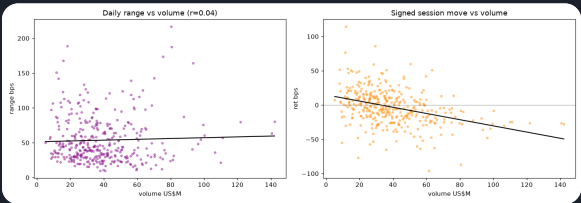
3 · VOLUME PROFILE — WHERE THE MARKET TRADED

US\$ volume at each ~1-colón level



Point of Control 507 ¢; the 504–509 band holds ~40% of lifetime volume. 70% value area 453–510 ¢ — current price sits at the bottom edge, i.e. price-discovery territory, not a range to fade.

Volume vs daily range (left) and signed move (right)



Volume does NOT widen the range ($r \approx 0.04$). It drives DIRECTION: the right panel's clear negative slope ($-0.45 \text{ bps}/\$M$, $p \approx 1e-13$) is the order-flow fingerprint — size = USD supply = colón strength.

4 · TRADING SIGNALS (GROSS OF COSTS)

SIGNAL	RESULT	VERDICT
Order-flow continuation short USD after heavy days, long after light	+7.0 bps/day · 59% hit · Sharpe $\approx$ 4.4	Best clean edge — works even in the flat regime
Overnight momentum follow a $>1\sigma$ session move into the next session	+17 bps/trade · 73% hit · Sharpe $\approx$ 8.4	Strong but fewer trades · same trend driver
Fade the opening gap buy/sell the gap vs prior VWAP back to the close	63% hit · Sharpe $\approx$ 6	Real (~80% retrace) but Sharpe inflated by shared-open noise

5d/20d MA crossover
classic trend-follow on the
VWAP

−1.3 bps/day · Sharpe $\approx$ −0.8

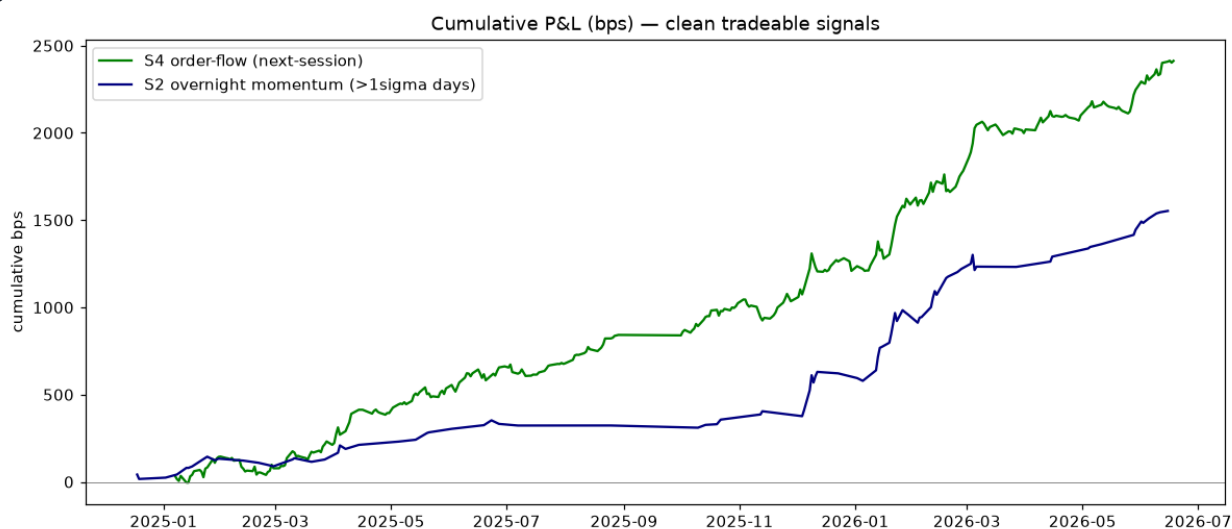
Whipsaws — the edge lives
at the 1-day horizon

Day-of-week bias
directional bias by weekday

Wed weakest ($t \approx -1.9$)

Suggestive only, not
significant

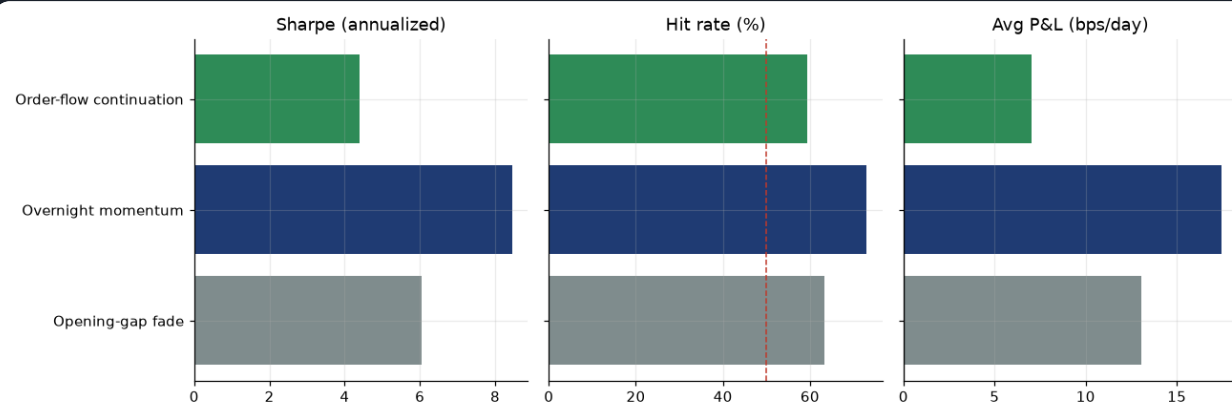
Cumulative P&L of the two clean tradeable signals



Both grind upward through the flat regime (Mar–Sep 2025) and accelerate into the trend — the order-flow signal (green) has standalone value beyond simply riding the appreciation.

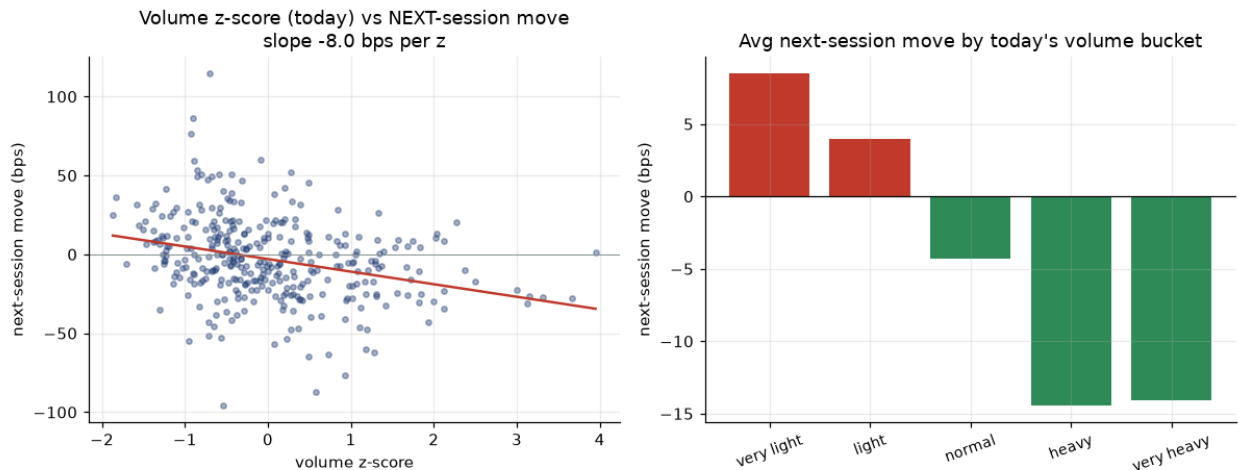
5 · STRATEGY DEEP-DIVE

Signals side-by-side — Sharpe, hit rate, average P&L



The order-flow and momentum signals clear 50% hit comfortably; the opening-gap fade is shown for completeness but its headline stats are inflated by the shared-open artifact.

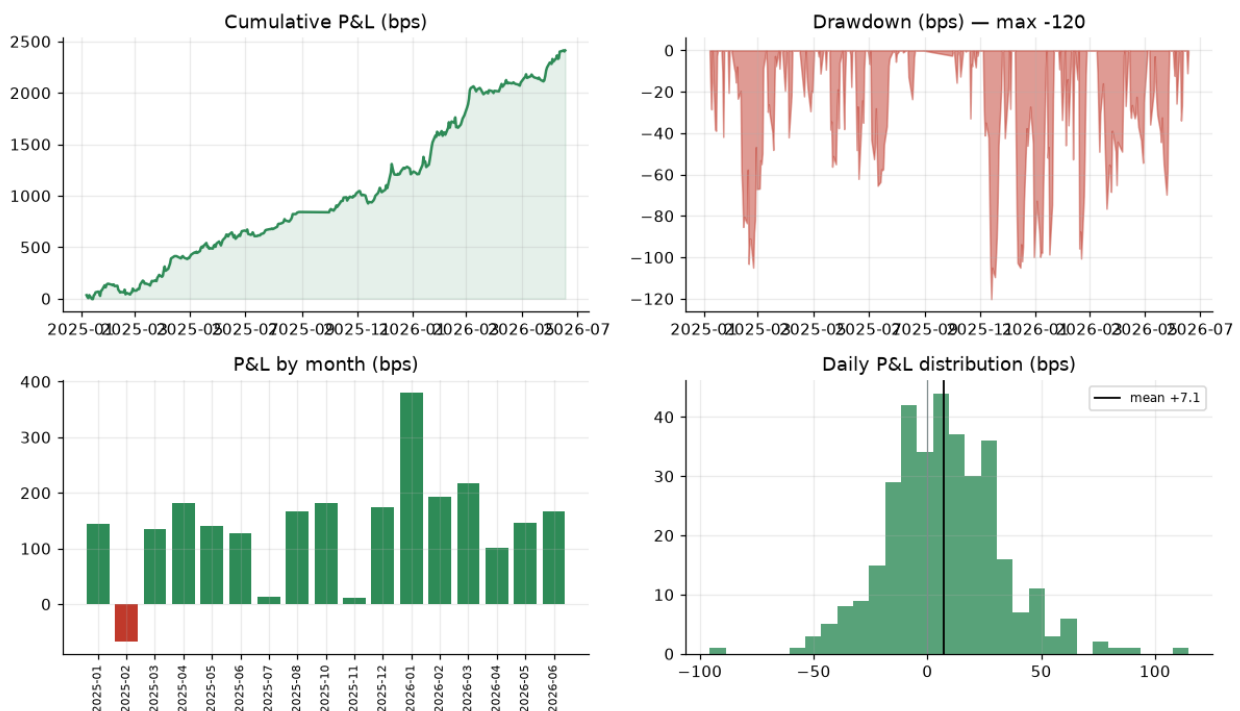
Why the order-flow edge exists



Left: today's volume z-score vs the NEXT session's move — a clean negative slope, so heavy days keep pushing the colón stronger tomorrow. Right: bucketed, the gradient is monotonic — very-heavy days → next-session appreciation, very-light days → depreciation.

Order-flow continuation — full tearsheet

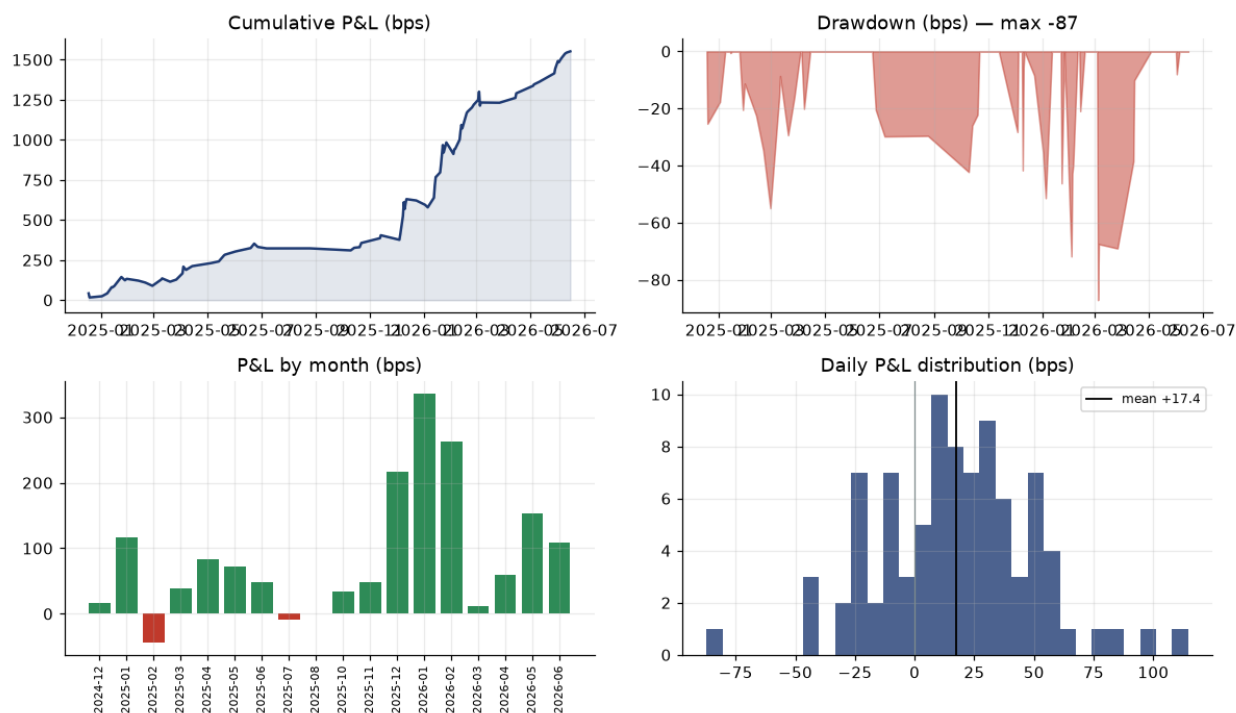
Order-flow continuation — strategy tearsheet (Sharpe 4.41 · 59% hit · +7.1 bps/day · n=342)



Equity, drawdown, monthly P&L, and the daily-P&L distribution. Smooth equity, only one losing month (Feb-25), shallow drawdowns, and a right-skewed daily distribution.

Overnight momentum — full tearsheet

Overnight momentum — strategy tearsheet (Sharpe 8.45 · 73% hit · +17.4 bps/day · n=89)



Fewer trades (only $>1\sigma$ session moves qualify) but a higher per-trade edge; the same trend/order-flow driver underlies both strategies.

HOW TO TRADE IT

The dominant trade is **long colón / short USD** (the crawl), sized **up after heavy-volume sessions** and into the **first half of the month / Feb & Apr**, trimmed in **December**.

Execute **VWAP-to-VWAP**: the open anchors to a known prior reference, so you can transact near a predictable rate.

Why it should persist: structural USD oversupply (exports, tourism, FDI, possible BCCR reserve accumulation) shows up as volume, and the managed-float anchor bleeds that pressure out gradually rather than gapping — hence the autocorrelation.

Caveats. All P&L is gross (no MONEX spread / brokerage / settlement); the colón is not freely shortable for most participants → a directional/hedging edge, not arbitrage. Backtests are in-sample over one appreciation regime; a policy shift would break the crawl. The "best board offers" block in the export was empty, so VWAP is the price proxy throughout.

